Exp. No: 10 PHOTODIODE RESPONSE AND SENSITIVITY ANALYSIS USING OPTISYSTEM

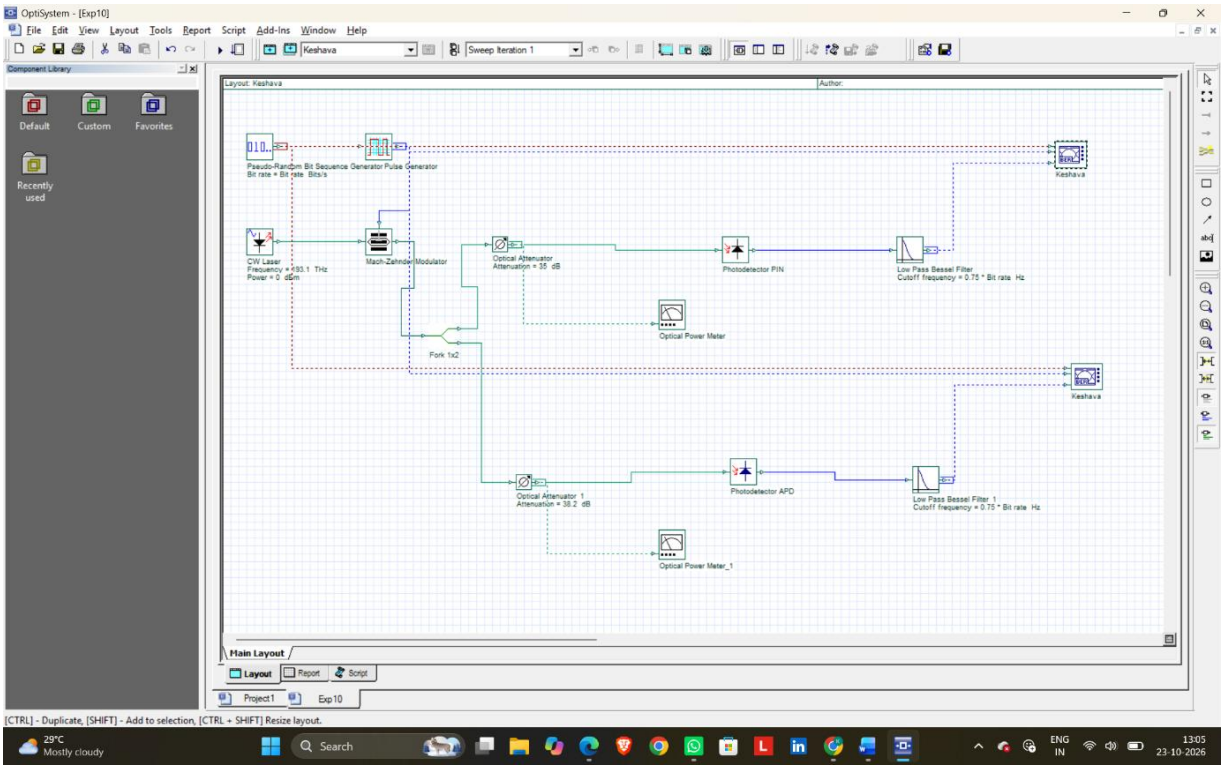
OBJECTIVES:

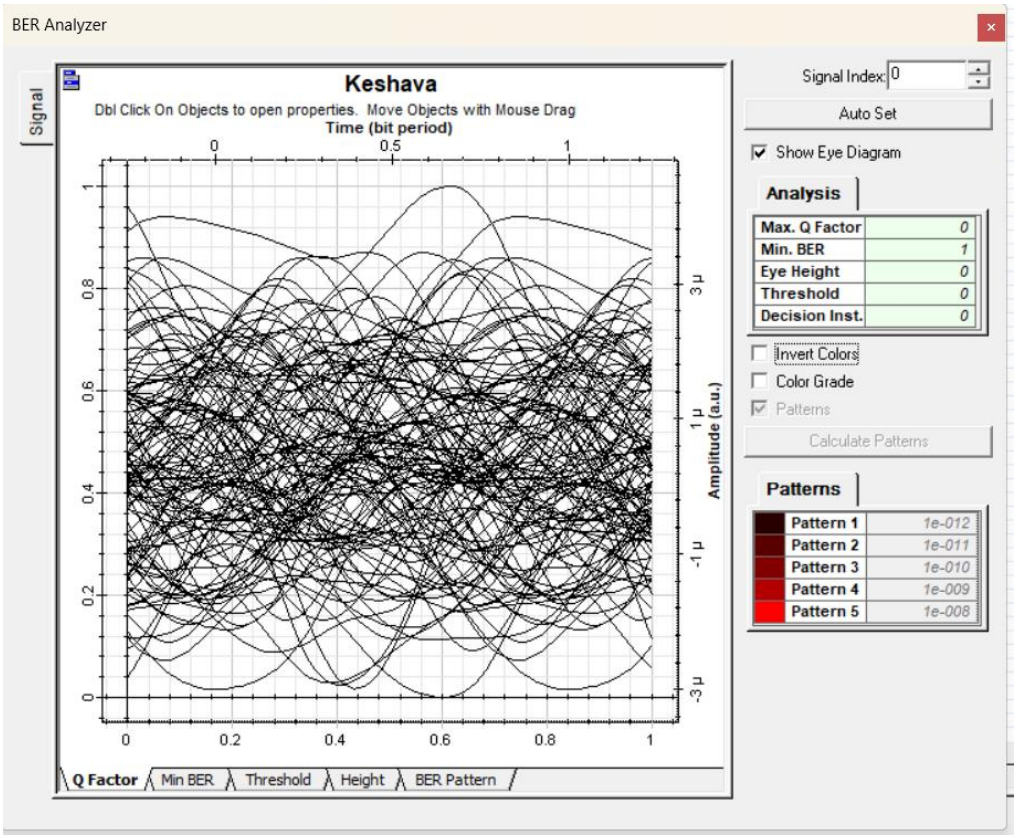
1. **Design and simulate an optical receiver using PIN and APD photodetectors in OptiSystem.**
2. **Observe the photocurrent generated for a given received optical power and determine the responsivity of the detectors.**
3. **Determine the receiver sensitivity (minimum received power), bandwidth and the variation of SNR and BER with received power.**
4. **Compare the PIN and APD receivers and evaluate photodetector performance.**

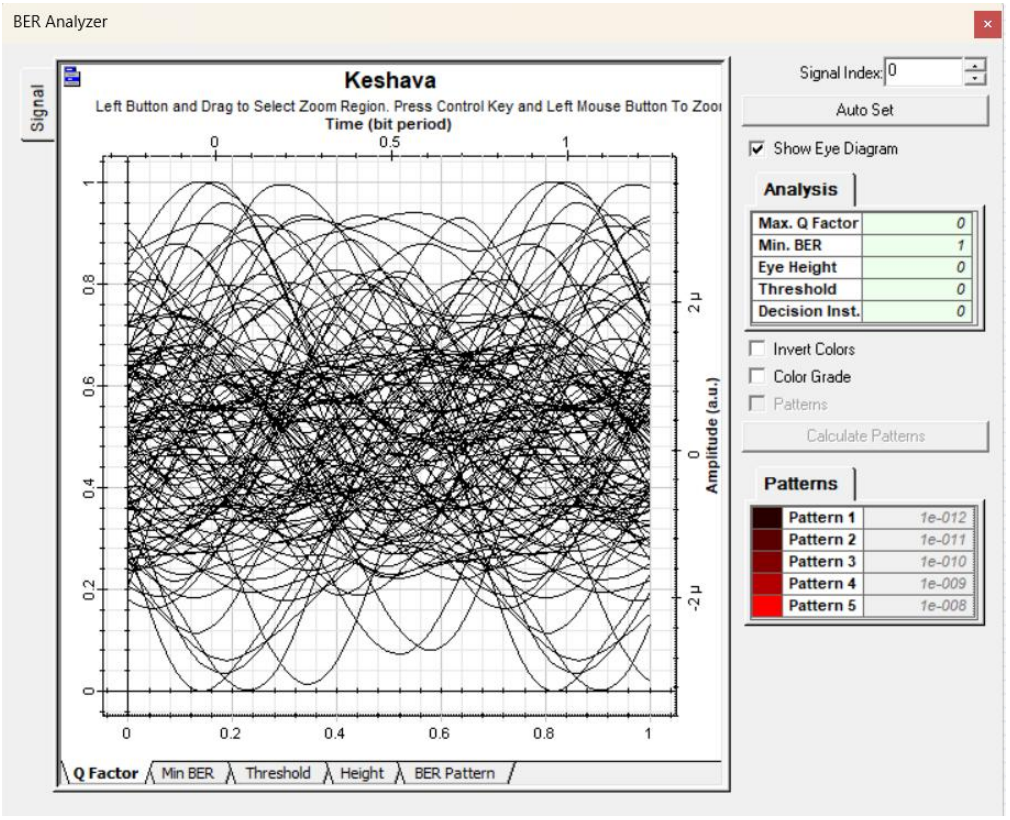
SOFTWARE REQUIRED:

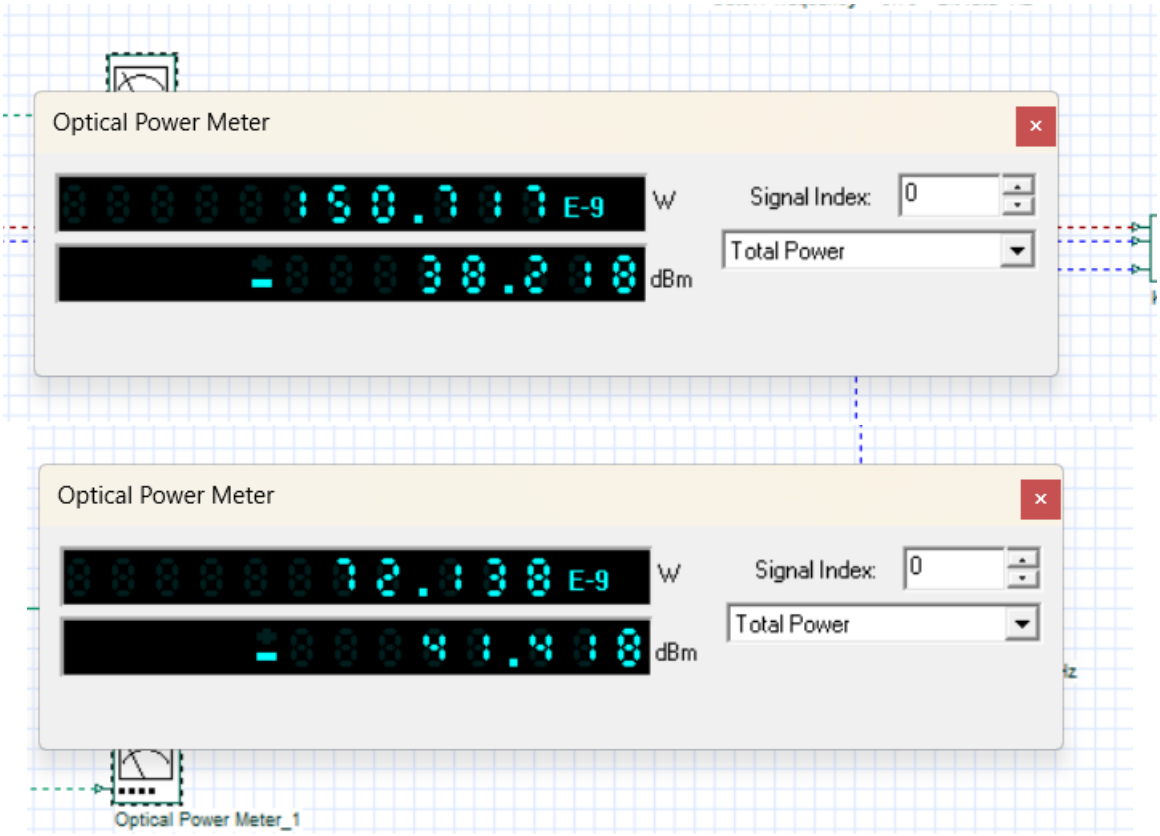
- **OptiSystem software**

BLOCK DIAGRAM:




BER Analyzer
Keshava
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal
Amplitude (a.u.)
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 0
Min. BER 1
Eye Height 0
Threshold 0
Decision Inst. 0
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern


BER Analyzer
Keshava
Left Button and Drag to Select Zoom Region. Press Control Key and Left Mouse Button To Zoo
Time (bit period)
Signal
Amplitude (a.u.)
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 0
Min. BER 1
Eye Height 0
Threshold 0
Decision Inst. 0
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern


Optical Power Meter
150.717 E-9
W
-30.218
dBm
Signal Index:
0
Total Power
Optical Power Meter
12.138 E-9
W
-41.418
dBm
Signal Index:
0
Total Power
Optical Power Meter_1

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improve-ment – 1	Weight-age
Design the OptiSystem PIN/APD receiver circuit with correct components and parameters.	Circuit Setup and Parameter Definition	Builds the complete circuit (laser, modulator, Fork 1×2, attenuators, PIN/APD, filters, power meters, BER analyzers) with all parameters correct (193.1 THz, 0 dBm, 35/38.2 dB, 0.75 × bit rate), independently.	Builds the circuit with minor connection or parameter errors and limited assistance.	Builds only part of the circuit or needs considerable assistance; has unit/parameter errors (e.g., dB vs dBm).	Unable to build the circuit or set the parameters.	25
Run the simulation and record received power, Q factor and BER for both receivers.	Simulation Execution and Data Acquisition	Runs the simulation and records received power (W and dBm), eye diagram, Q factor, BER and the attenuation sweep correctly for PIN and APD.	Records all quantities with minor errors or omissions.	Records some quantities with noticeable errors or inconsistencies.	Unable to run the simulation or record the values.	25
Obtain eye diagrams and BER/SNR plots; determine responsivity, sensitivity and bandwidth.	Graph Plotting and Sensitivity Determination	Plots accurate eye diagrams and BER (semi-log) / SNR curves with labelled axes and a BER = 10^{-9} line; sensitivity within ±1 dB, responsivity and bandwidth within 10% of the reference.	Plots the curves with minor errors; sensitivity, responsivity and bandwidth identified with small deviation.	Plots partial or poorly scaled curves; sensitivity, responsivity or bandwidth approximate.	Unable to plot the curves or determine the parameters.	25
Interpret photocurrent–power relation, SNR/BER variation and PIN vs APD performance.	Analysis and Engineering Interpretation	Correctly explains linear photocurrent, noise-limited regions, SNR/BER trend and the APD gain advantage, and links results to detector selection.	Explains the trends with minor errors and reasonable interpretation.	Performs basic interpretation with noticeable errors or limited justification.	Unable to interpret the results.	25

CONCLUSION:

The experiment shows how the photodetector converts received optical power into photocurrent and how this determines receiver performance. The following observations were made:

• PIN receiver: with −38.2 dBm at the detector the eye is open; photocurrent is proportional to optical power with responsivity of about 1 A/W.

• APD receiver: with 3.2 dB less power (−41.4 dBm) the eye is still open and its amplitude is larger, since avalanche gain (≈ 3) increases the effective responsivity and sensitivity.

• Power dependence: below the sensitivity level SNR falls and BER approaches 0.5; above it the SNR and Q factor rise rapidly and the link is error-free.

• Bandwidth: the photodiode bandwidth is much larger than the bit rate, so the receiver bandwidth is set by the low pass filter at 0.75 × bit rate.

The APD gives be er sensitivity at the cost of added avalanche noise and higher bias, whereas the PIN photodiode is simpler and less noisy at high power. These results help in selecting the detector for a given power budget.

Reflection:

I learned how to analyze photodiode response and sensitivity using OptiSystem. The experiment helped me understand the relationship between received optical power, photodetector sensitivity, and signal quality.